DESK HOLOGRAM · THE BUILD, STEP BY STEP

Two days at the bench

Rev B of 2026-11-07, written for Monday 2026-11-09 and Tuesday 2026-11-10. Every step says what to take out, what to do, and what counts as a pass; nothing here is optional. The pins, nets, parts and dimensions are imported from the build sheet's generator, so they cannot drift from it. The sheets themselves: Desk Hologram Build; the physics: Desk Hologram Physics. Nothing built; nothing measured.

What you end up with

A 72 × 67 × 77 mm desk object, about 354 g: the display module in a brass-topped ivory skin on four stainless legs, the 50 mm cube on the glass, a 4 mm slot of amber light under the skirt. Press the button, speak, and the answer floats in the glass. Monday is the electronics and the skin's paint; Tuesday is the lining, the legs, the skin and the cube.

BEFORE THE IRON GOES ON, THE THREE RULES

Design and colour first: the skin is why the first hour is card and brass, not solder. Meter only what matters: polarity, isolation and the supply rail, nothing a supplier already specified. Component names, not product codes; the link is on the word.

The pins, for the bench

The approved wiring sheet in one table: the controller's pin, its job, where the wire goes. Step 6 turns it into seventeen cut lengths.

PIN	JOB	GOES TO
3V3	3.3 V out of the board's regulator	display pin 2 (3V) · microphone VDD
GND	ground	display pin 1 (GND) · microphone GND and L/R · button · display pin 20 (frame) optional
GPIO6	SPI clock	display pin 7 (D0)
GPIO7	SPI data	display pin 8 (D1)
GPIO5	chip select	display pin 15 (CS)
GPIO20	data / command	display pin 4 (DC)
GPIO21	reset	display pin 16 (RESET)
GPIO1	I2S word select	microphone WS
GPIO3	I2S bit clock	microphone SCK
GPIO4	I2S data in	microphone SD
GPIO10	push to talk, active low, internal pull-up	button, other leg to GND
GPIO0	LED filament, PWM-dimmed through 68 Ω	68 Ω, then the filament's + lead; its − lead to GND
GPIO2 · 8 · 9 · 5V	nothing	strapping pins, the BOOT button and the 5 V pin stay empty

STEP 0 Order the brass, meet the cube, caliper the module

Sat 11-07 to Sun 11-08

DO

1. Order one sheet of brass, 0.005 in (K&S 250, a 6-pack) and one tube of gel super glue (Loctite Ultra Gel Control, the one in your Amazon Saved for later). They are the only purchases; the rover's list carries both lines, so one order serves both builds. If it has not landed by Monday, Monday runs steps 3 to 9 without it and the skin follows the day it comes.
2. Sunday: the cube arrives. Pick it up when the Todoist reminder fires. Incoming check: no chips on the edges, the diagonal a faint plane inside the glass when you look through the front face. Back in its wrap until Tuesday: glass on glass scratches.
3. Caliper the spare display module and write five numbers into `D` at the top of `docs/pages/gen/build_page.py`: the board's width and depth (the record says 72 × 43), a mounting hole's centre from its two edges (3), the glass outline (60.5 × 37) and the margin between the glass and the back edge, where the header row is. Rerun `python3 docs/pages/gen/build_page.py --pdf` and `python3 docs/pages/gen/build_day_page.py --pdf`; the pattern page redraws itself.
4. Print, Letter, black, Actual size: the two sheets of the build PDF, and this PDF's pattern page on a sheet of the cardstock (the DeskJet takes 65 lb card). Check the 100 mm bar with the calipers before cutting anything.
5. Charge nothing, buy nothing else. Everything below is in the bins.

PASS

The cube is in the flat, the brass and the glue are on their way, the five numbers are in the generator, and the printed bar measures 100 mm.

MONDAY'S SHAPE

The skin is cut and painted first, so the paint dries while the wiring happens. The display's two links move before anything else is soldered to it. The controller is flashed bare. Then the seventeen wires, first light, ears, lamp. Tuesday: Kapton, legs, skin on, cube on. About five hours Monday, two Tuesday.

STEP 1 The bench

Mon 11-09, first thing

TAKE OUT

- the spare 2.42" display kit: the module, its 20-pin header strip and the 220 µF from the bag; the level shifter stays in the bag
- one ESP32-C3 SuperMini, a fresh one, gated in step 5
- one I2S microphone

- one 6 mm tactile button and a white cap from the QTEATAK kit
- the 68 Ω resistor; one 100 nF from the ceramic kit
- 30 AWG silicone wire: black, red, white
- Kapton, white heat-shrink (2 and 3 mm), the gel super glue (bought with the brass; the hot-glue gun is the only glue owned, and it lets go of cold metal), 63/37 solder, flux, wick, baking soda, 91 % alcohol, swabs
- one LED filament of the ten
- the brass sheet when it has landed; the canvas-white spray; sandpaper 400, 800, 1500, 3000; the cardstock sheet with the pattern printed on it; the craft knife and mat; scissors; the jeweller's saw; the chain-nose and round-nose pliers; the diamond needle files; the flush cutters; the calipers
- one 1.5 mm stainless rod, 150 mm
- the multimeter, the magnifying lamp, the safety glasses, a clean cloth for the glass
- the laptop with ESP-IDF (the Pocket AI's install), a USB-C data cable, the 2 A adapter; the desktop awake for the voice server
- the cube, Tuesday

DO

1. Lay it out left to right in the order of the steps. The module goes glass-down on the cloth whenever it is on its back.
2. Print the wire table below, or keep this page open at the bench.

PASS

Everything on the list is on the bench before the iron is on.

STEP 2 The skin, first thing, so the paint dries while you wire

`Mon, 1 h, then it dries`

TAKE OUT

- the cardstock with the pattern on it, the brass sheet and an offcut of it, the knife and mat, scissors, pliers, needle files, 400 / 800 / 1500 / 3000, alcohol, Kapton, the iron with 63/37 and flux, baking soda, the spray can

DO

1. Card first. Cut the printed pattern out of the card: the outline, the window (56.5 × 33 as drawn) with its four 45° relief cuts out to the 60.5 × 37 fold line, the four Ø2.5 leg holes, the 7 × 7 button hole, the Ø3 mic hole in the front flap, the 10 × 5 slot in the back flap. Fold the flaps down 16, the window's 2 mm lip up, the tabs round behind the front and back flaps.
2. Try it on the module: the glass in the window with the lip up against its edge and no green showing; the leg holes over the mounting holes; the slot where the controller's USB-C will sit (back edge, 23 mm right of centre); the mic hole at the front left. Mark every correction on the card. The card is the pattern you may cut twice; the brass is cut once.

3. Transfer: lay the corrected card on the brass sheet along its length (104 along the 254), tape it down with Kapton, scribe round everything with the knife's point. Cut the outline with scissors, the window and the slot with the knife on the mat in several light passes (the relief cuts last), the round holes by piercing with the knife's point and opening with the round needle file to size.
4. Deburr every edge with 400 grit. A brass edge cuts a wire, or a hand.
5. Fold: each flap over a steel rule or the bench edge, pressed flat with the chain-nose pliers; the 2 mm lip with the chain-nose pliers, a little at a time along all four sides; the tabs round behind the front and back flaps. Try the tray on the module again.
6. Corners: a dab of 63/37 with flux on the inside of each tab, the iron for a second, done. Then baking soda and water on every joint, rinse, alcohol, dry: flux left under paint blisters it.
7. Finish: 400 grit over the outside of the four flaps and an alcohol wipe. The top frame and the lip: 800, 1500, 3000 wet, then alcohol, until bright. Mask the top frame and the lip with Kapton cut exactly at the fold line; mask nothing else.
8. Paint outdoors or at an open window, never near the iron. Spray the brass offcut first. Two light coats of gloss canvas white on the four flaps, twenty minutes apart. Hang the tray by a wire through a leg hole. Peel the Kapton mask after an hour, while the paint is soft, pulling along the fold, never across it. Leave it overnight before the lining goes in.

PASS

The card tray fits the module with no green at the glass. The brass tray sits on the module without rocking, the button hole over the button's spot and the leg holes over the module's holes. After an hour the offcut's paint does not lift under a fingernail.

IF NOT

- Paint lifts on the offcut: the surface was not keyed or not clean. 400 again, alcohol, a thinner first coat.
- The lip will not fold cleanly: the relief cuts stop short of the fold line. Extend them.
- The tray rocks: a flap is short of 90°. Press it on the bench edge.

BRASS IS NOT A CONDUCTOR HERE

Kapton goes on every inside face on Tuesday, and the meter proves it before power. Keep the lip clear of the glass's flex cable at the back edge; if the flex sits proud there, notch the lip over it.

STEP 3 The display's two links: parallel to SPI

Mon, 20 min

TAKE OUT

- the module, the iron, wick, flux, alcohol, the lamp, the phone camera

DO

1. Photograph the back before touching it, as Cyber Dawn did (`cyber_clock_v2/docs/photos/`). Find the BS1 block (R2 over R3) and the BS2 block (R4 over R5), left of centre below the header row. As shipped the parts sit in the upper "(1)" positions, R2 and R4: that is 8080 parallel, and the glass stays dark on SPI.
2. Take the part off R2: wet both ends with a little fresh solder, slide it off with the tip. Same on R4. Wick the pads clean.
3. Bridge R3 with one solder blob across its two pads; the same on R5. That is BS1 = 0 and BS2 = 0: 4-wire SPI.
4. Alcohol over the area; look under the lamp.

PASS

Under the lamp, R2 and R4 are bare pads and R3 and R5 each carry one bright bridge; nothing else on the board has moved.

IF NOT

- A bridge reaches into the neighbour row: wick it off and start again.
- A pad lifts: stop. The other board of the pair is Cyber Dawn's and is already reworked; this one has to be got right, slowly.

WHY

The panel's controller reads BS1 and BS2 at power-up to pick its bus. Cyber Dawn's glass stayed dark through every test on 2026-11-07 until these links were moved; after that the all-pixels-on command lit it first time.

STEP 4 Header and the two capacitors

Mon, 20 min

TAKE OUT

- the module, the 20-pin header, the 220 µF, the 100 nF, the iron, Kapton, the meter

DO

1. A strip of Kapton over the edge of the glass nearest the header row, so the iron never meets glass.
2. The header on the back of the board, plastic body against the board, pins pointing down, away from the glass. Tack pin 1 and pin 20 from the front face, check it stands square, then solder the other eighteen: small bright joints, the iron angled away from the glass.
3. The 220 µF across pins 1 and 2 on the back, lying flat along the header: its striped lead (−) to pin 1 (GND), the other to pin 2 (3V). The 100 nF across the same two pins beside it. Trim the leads with the flush cutters.
4. The one meter check allowed here: pins 1 and 2 not shorted. A chirp that stops is the 220 µF charging.

PASS

The header stands square with twenty bright joints; the meter chirps and stops between pins 1 and 2.

IF NOT

- A steady beep between pins 1 and 2: the electrolytic is in backwards, or a bridge. The lamp will show which.

STEP 5 The controller alone: gate, flash, join

Mon, 20 min

TAKE OUT

- one SuperMini, the USB-C data cable, the laptop; the desktop for the server

DO

1. On the desktop, start the voice server the way the first Pocket AI's is started, bound to the LAN. `POCKET_TOKEN` must be the same string as `CONFIG_POCKET_TOKEN` in the firmware's `sdkconfig.secrets`, or the link never comes up:

```
cd pocket_ai_assistant/character/server
POCKET_BRAIN=ollama POCKET_TTS=none POCKET_TOKEN="<the CONFIG_POCKET_TOKEN string from sdkconfig.secrets>" \
  .venv-speech/bin/python -m pocket_server --host 0.0.0.0
```

2. Plug the bare SuperMini in and gate it: the board must name an ESP32-C3, or it goes back in the bag.

```
esptool --port /dev/ttyACM0 flash_id
```

3. Build and flash (ESP-IDF 6.0.2 is already installed under the Pocket AI's firmware):

```
cd desk_hologram/firmware
source ../../pocket_ai_assistant/firmware/.work/esp-idf/export.sh
cp ../../pocket_ai_assistant/character/firmware/sdkconfig.secrets .
SDKCONFIG_DEFAULTS="sdkconfig.defaults;sdkconfig.secrets" idf.py set-target esp32c3
idf.py build
idf.py -p /dev/ttyACM0 flash monitor
```

4. Watch the monitor. With nothing wired yet, the log is the whole picture: the Wi-Fi link, then `server link` up.

PASS

The log shows the Wi-Fi link and the server link up, and no reset over a full minute.

IF NOT

- SSID empty in the log: the secrets file was not in place before `set-target`. Copy it and build again.
- Wi-Fi up, no server: the server is bound to 127.0.0.1 (it wants `--host 0.0.0.0`), the desktop's firewall, or the token differs.

STEP 6 The seventeen wires

Mon, 90 min

TAKE OUT

- the module with its header, the flashed SuperMini, the microphone, the button and cap, the 68 Ω, the filament, wire, Kapton, heat-shrink, the iron, the meter

DO

1. Cut the wires to the table below, all with slack, and trim at the far end. Red for 3.3 V, black for ground, white for signals: all of it hides under the skin.
2. Display first: wires 1 to 7 onto header pins 1, 2, 4, 7, 8, 15 and 16. The pins point down, so the wires run along the underside toward the back right.
3. Tape the controller under the module at the back right with Kapton, components down, its USB-C flush with the back edge and 23 mm right of centre (the slot in the skin is drawn there). Solder the seven display wires to 3V3, GND, GPIO20, GPIO6, GPIO7, GPIO5 and GPIO21.
4. The microphone: on its breakout, wire 8, the 10 mm black jumper from L/R to its own GND pad. Then wires 9 to 13 to the breakout. Tape it under the module at the front left, 24 mm left of centre, its port toward the front edge; solder VDD to 3V3 (that pin's second wire), GND to GND, WS to GPIO1, SCK to GPIO3, SD to GPIO4.
5. The button: wires 14 and 15 to its legs, heat-shrink on each. A dab of the gel super glue under the button and set it on top of the module at the front right, 27.5 mm right of centre and 3.5 mm in from the front edge, cap on; the wires go over the edge and under. Leg 1 to GPIO10, leg 2 to GND.
6. The filament: the 68 Ω in line with the + lead, heat-shrunk; wire 16 from the resistor to GPIO0; wire 17 from the − lead to GND. Tape the filament flat under the front edge beside the microphone, glowing down, leads toward the controller.
7. Grounds: the display's and the button's go straight onto the controller's GND pin; twist the microphone's ground and the filament's − together into the same joint. Four ends on one pad: tin it first, add them one at a time.
8. Kapton over every wire run so nothing can lift, and over the back of the controller.
9. Meter before any power: every signal wire beeps to its controller pin and to no neighbour; 3V3 to GND not shorted; display pins 1 and 2 as in step 4.

THE WIRE TABLE

#	NET	FROM	TO	COLOUR	CUT, MM
1	GND	display pin 1 (GND)	controller GND	black	45

2	3V3	display pin 2 (3V)	controller 3V3	red	45
3	DC	display pin 4 (DC)	GPIO20	white	40
4	SCLK	display pin 7 (D0)	GPIO6	white	40
5	MOSI	display pin 8 (D1)	GPIO7	white	40
6	CS	display pin 15 (CS)	GPIO5	white	40
7	RST	display pin 16 (RES)	GPIO21	white	40
8	GND	microphone L/R	the microphone's own GND pad (same net, on the breakout)	black	10
9	GND	microphone GND	controller GND	black	75
10	3V3	microphone VDD	controller 3V3 (the pin's second wire)	red	75
11	WS	microphone WS	GPIO1	white	75
12	SCK	microphone SCK	GPIO3	white	75
13	SD	microphone SD	GPIO4	white	75
14	BTN	button, leg 1	GPIO10	white	55
15	GND	button, leg 2	controller GND	black	55
16	LED	filament + lead, through the 68 Ω	GPIO0	white	50
17	GND	filament − lead	controller GND	black	50

PASS

17 wires, 2 capacitors, 1 resistor. Nothing stands proud of the underside by more than the header's 9 mm, so it all fits inside the 16 mm skirt. The meter checks pass.

IF NOT

- A beep between neighbours: one strand of 30 AWG bridging two pins. Alcohol, a brush, the lamp.

THE RULES

Never feed the SuperMini's 5V pin. The display and the microphone take 3.3 V only, from the controller's 3V3 pin; never connect another source to that rail while USB is in. Neither the brass nor the legs carry a net. Keep flux, alcohol, glue and paint away from the microphone's port and the glass.

STEP 7 First light

Mon

TAKE OUT

- the wired module, the cable, the adapter

DO

1. USB in. The firmware pulls the lamp pin low, resets the display and draws its state word: `wifi`, then `server`, then the character's name as it connects.
2. Look at the bare glass from above: the text reads backwards. That is right; the cube turns it round.

PASS

Text on the glass within five seconds of power, mirrored, steady.

IF NOT

- Dark: step 3's links are the first suspect (Cyber Dawn's lesson), then DC on GPIO20 and RESET on GPIO21 swapped, then 3.3 V at display pin 2 (meter the rail: a supply voltage is one of the three things that gets metered).
- Flicker or a faint picture: the 220 µF is not across pins 1 and 2.

STEP 8 Ears

Mon

TAKE OUT

- the server running from step 5

DO

1. Hold the button: `listening` on the glass, the lamp brighter. Say one sentence. Release: `thinking`, the lamp breathing; then your words flash up as the desktop heard them, then the answer, paged.
2. Ask three things in a row. Watch the server's log for `heard nothing`.

PASS

Your words come back as the desktop heard them, then an answer, three times out of three.

IF NOT

- `heard nothing` or silence: L/R not grounded, SD not on GPIO4, WS and SCK swapped, or the port taped over.
- Words but no answer: Ollama is not running on the desktop (127.0.0.1:11434).

STEP 9 The lamp

`Mon, last thing`

DO

1. At rest, a faint amber under the front edge. While the button is held, brighter. While the desktop thinks, a slow breath.
2. USB out. The module on its back on the cloth. The skin stays hanging to dry.

PASS

The three levels, by eye.

IF NOT

- Dark: the filament is in backwards (the + lead is the one through the resistor to GPIO0). Swap the leads.
- Full brightness always: GPIO0 is not the lamp pin in menuconfig.

STEP 10 Lining, legs, skin on

`Tue 11-10, 90 min`

TAKE OUT

- the painted skin, Kapton, one stainless rod, the jeweller's saw, the pliers, 400 grit, white heat-shrink, the gel super glue, the heat gun, the meter

DO

1. Kapton on every inside face of the skin: under the top frame, inside the four flaps, inside the lip. Trim it back at the holes.
2. Legs: four at 37 mm from the 150 mm rod with the jeweller's saw; deburr both ends. Bend each once, 22 mm from one end, into an L with the chain-nose pliers, the 15 mm foot square. White heat-shrink on the shank where it passes the hole and the skin; a cap of heat-shrink on the foot's tip.
3. Skin on: the USB-C plug through the back slot first, then the tray down over the module, the button up through its hole, the glass up through the window with the lip against it, the mic hole in front of the port.

4. Legs in: each leg up from below through the skin's hole and the module's hole until the shank's top is flush with the glass-side face, the foot pointing straight out, front legs forward and back legs back. A drop of the gel super glue at each hole on top and one where the leg passes the skin. Stand it on the desk and square it while the glue grabs. An hour to cure; a day before it carries the cube.
5. Isolation, before power: rod to GND open, rod to 3V open; brass to GND open, brass to 3V open, brass to each leg open.

PASS

Level, no rock, 20 mm clear under the module and a 4 mm slot of light all round; the five isolation checks open.

IF NOT

- A beep from the brass: a bare brass edge is touching a header pin. Lift the skin; more Kapton there.

STEP 11 The cube

Tue

TAKE OUT

- the cube, the black card if you have one, the phone or the webcam page

DO

1. USB in; the glass shows its word. Unwrap the cube and set it on the glass, front face toward you, centred on the pixels. Turn it until the text reads the right way round and stands upright inside.
2. The picture stands at the back face, 11 mm up. If the room shows through it, a black card behind the cube.
3. A photo for the record, and three numbers: mass, height, the slot.

PASS

The answer floats in the glass, right way round, from a seat level with the cube to 30° above it. Built.

THEN WRITE IT DOWN

The cube's and the brass sheet's pick-up dates into `docs/INVENTORY.md`; "built 2026-11-10" and the photo into the README; the calipered numbers into the generator if they moved anything.

The meter, all of it

The only measurements in the build. Everything else is a supplier's number and is not metered.

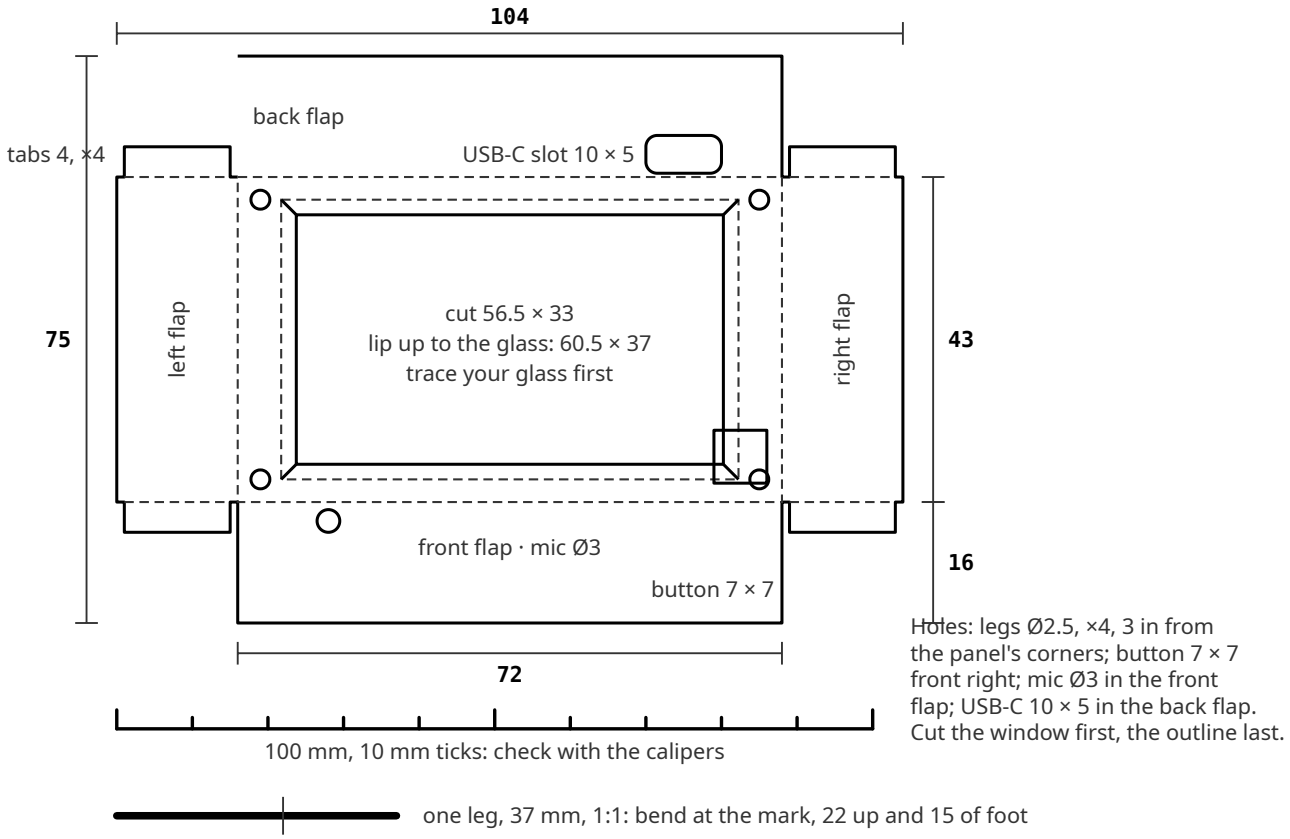
- [ ] Display header, before the controller is powered: pins 1 and 2 (GND and 3V) not shorted. A chirp that stops is the 220 µF charging.
- [ ] Each of the five display signal wires beeps to its controller pin and to nothing next to it.
- [ ] The legs' heat-shrink sleeves isolate the stainless from the module's pads: rod to GND and rod to 3V open.
- [ ] The skin is not a conductor, before power and with it fitted: brass to GND open, brass to 3V open, brass to each leg open. The Kapton lining is what keeps them so; a beep means a lifted edge. Everything else is a supplier spec and is not metered.

The skin pattern, 1:1

Built by `docs/pages/gen/build_day_page.py`, which also writes `docs/BUILD_DAY.md` and the PDFs; the wire table is checked against the sheet's nets and pins every time it runs. Wire lengths are read off the top view with slack; the caliper step on Sunday is what makes the pattern true.

Skin pattern, 1:1

Print on the cardstock, Letter, black, Actual size. Solid = cut, dashed = fold. The bar must measure 100 mm.



Holes: legs Ø2.5, ×4, 3 in from the panel's corners; button 7 × 7 front right; mic Ø3 in the front flap; USB-C 10 × 5 in the back flap. Cut the window first, the outline last.

Rev B, 2026-11-07. Cyber Dawn's numbers (72 × 43, holes 3 in): caliper the spare, rerun the generator, then cut.